

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T39 · Standard and Compound Units

With Answers

Questions with full worked solutions

■ 6 pages

FAMILY MAP

Standard Compound Units

F01

Select and convert standard metric measures

1 Q | 2 marks

F02

Read time and measuring instruments

1 Q | 6 marks

2 Change 32.4 m^3 into cm^3

..... cm^3

Worked solution

January 2020 | Question 2 | 2 marks

Family: Select and convert standard metric measures | Variant: Convert cubic metric units and capacity

METHOD / WORKING

$$1. 1m^3 = 100^3 = 1,000,000cm^3.$$

$$2. 32.4 \times 1,000,000 = 32,400,000cm^3.$$

FINAL ANSWER

$$32,400,000cm^3$$

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

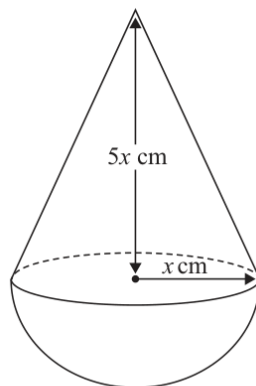


Diagram **NOT**
accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.



..... g/cm³

(Total for Question 26 is 6 marks)

Worked solution

November 2025 | Question 26 | 6 marks

Family: Read time and measuring instruments | Variant: Make a sensible estimate of a measure

METHOD / WORKING

Worked solution
November 2025 Question 26 6 marks
Family: Read time and measuring instruments Variant: Make a sensible estimate of a measure
Method
Volume of cone:
$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi (5x)^2 (6x) = \frac{50}{3}\pi x^3$
Volume of hemisphere:
$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi (6)^3 = 48\pi$
Total volume:
$\frac{50}{3}\pi x^3 + 48\pi = 504\pi$
$\frac{50}{3}\pi x^3 = 504\pi$
$x^3 = 29.16$
$x = 3$
Cone volume:
$\frac{1}{3}\pi (5)^2 (6) = 50\pi$
Mass of copper cone:
$50\pi \times 7.2 = 360\pi$
Mass of hemisphere:
$48\pi - 360\pi = -312\pi$
Hemisphere volume:
$\frac{2}{3}\pi (6)^3 = 48\pi$
Density of the hemisphere:
$(1512\pi) / (48\pi) = 31.5$
Answer: 10.5 g/cm^3
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER